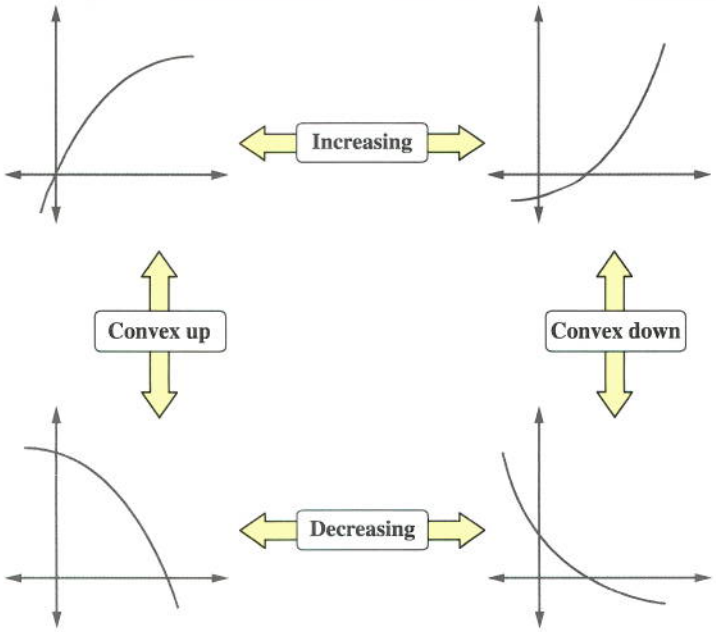
Unit 2

Lesson Five

Curve studying

⊙ Our study includes the 3rd degree polynomial or less as $f(x) = a x^3 + b x^2 + c x + d$

The relation between the shape of the curves "Increasing or decreasing - convexity"



⊙ Steps of drawing a curve of function f (where f is a polynomial of 3rd degree or less) :

1. Determine the domain of the function then determine the symmetric of the function $\hat{f}$ where :

(a) $f(-x) = f(x)$ for every $x \in$ the domain

$\therefore$ The function is even and so the curve is symmetric about y-axis.

(b) $f(-x) = -f(x)$ for every $x \in$ the domain

$\therefore$ The function is odd and so the function is symmetric about origin point.

2. Find out $\hat{f}(x)$, $\hat{\hat{f}}(x)$

3. Use $\hat{f}(x)$ to determine :

(a) The increasing interval where $[\hat{f}(x) > 0]$, decreasing interval where $[\hat{f}(x) < 0]$

(b) The points of local maximum and local minimum (if exist) where $\hat{f}(x) = 0$

(notice that the function is differentiable) and the sign of $\hat{f}(x)$ changes before and after this point.

4. Use $f''(x)$ to determine :

(a) The intervals where the curve convex upwards where $[f''(x) < 0]$, the intervals where the curve convex downwards where $[f''(x) > 0]$

(b) The inflection point (if exist) where $f''(x) = 0$ (notice that the function is differentiable twice) and the sign of $f''(x)$ changes before and after the point.

5. Determine some assisting in sketching as :

(a) The point of intersection with x-axis

(b) The point of intersection with y-axis

(c) Some extra points by substitution by some values of x and get the corresponding values of $f(x)$

6. Arrange the points we get in a table and represent them graphically then join these points taking in consideration the following :

Sign of $f'(x)$, $f''(x)$	Properties of the curve of function f	Shape of the curve
$f'(x) > 0$, $f''(x) > 0$	Increasing , convex downwards	
$f'(x) > 0$, $f''(x) < 0$	Increasing , convex upwards	
$f'(x) < 0$, $f''(x) > 0$	Decreasing , convex downwards	
$f'(x) < 0$, $f''(x) < 0$	Decreasing , convex upwards	

Example 1

Sketch the curve of the function $f(x) = (x + 2)(x - 1)^2$

Solution

The function f is a polynomial , its domain is $\mathbb{R}$, and the function is neither odd nor even.

$\because f(x) = (x + 2)(x - 1)^2 = (x + 2)(x^2 - 2x + 1) = x^3 - 3x + 2$

, $f'(x) = 3x^2 - 3$, $f''(x) = 6x$

1. Put $f'(x) = 0$

$\therefore 3(x^2 - 1) = 0$

$\therefore x = 1$, $x = -1$

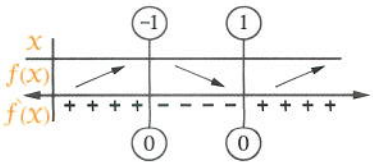
$\therefore 3x^2 - 3 = 0$

$\therefore 3(x - 1)(x + 1) = 0$

$\therefore$ The curve of the function is increasing

in $]-\infty , -1[$, $]1 , \infty[$

, the curve of the function is decreasing in $]-1 , 1[$



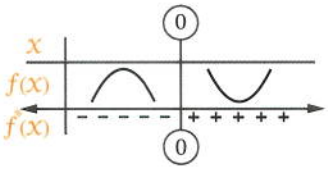
, $\because f(1) = 0 , f(-1) = 4$

$\therefore (1 , 0)$ is a local minimum point , $(-1 , 4)$ is a local maximum point.

2. Put $f''(x) = 0$ $\therefore 6x = 0$ $\therefore x = 0$

$\therefore$ The curve of the function is convex upwards in $]-\infty , 0[$

, the curve of the function is convex downwards in $]0 , \infty[$



, the inflection point at $x = 0$

, $\because f(0) = 2$ $\therefore (0 , 2)$ is an inflection point.

3. To find the point of intersection with x-axis , put $f(x) = 0$

$\therefore (x + 2)(x - 1)^2 = 0$

$\therefore x = -2$, $x = 1$

$\therefore$ The curve intersects the x-axis at

the points $(-2 , 0) , (1 , 0)$

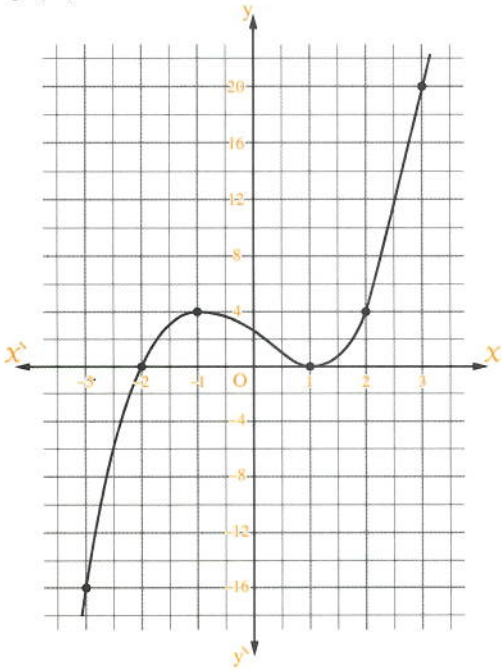
4. To find the point of intersection with y-axis

, put $x = 0$

, $\because f(0) = 2$

$\therefore$ The curve intersects the y-axis at

the point $(0 , 2)$



5. Find some assistant points

$\because f(3) = 20$, $f(-3) = -16$, $f(2) = 4$

$\therefore$ The curve passes through $(3 , 20) , (-3 , -16) , (2 , 4)$

6. Arrange all the previous points in the following table and represent them graphically :

x	-3	-2	-1	0	1	2	3
$f(x)$	-16	0	4	2	0	4	20

Example 2

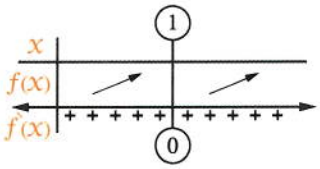
Sketch the curve of the function $f : f(x) = x^3 - 3x^2 + 3x + 7$

Solution

The function f is a polynomial function , its domain is $\mathbb{R}$, and the function is neither odd nor even.

$\because f(x) = x^3 - 3x^2 + 3x + 7$

$\therefore \dot{f}(x) = 3x^2 - 6x + 3$, $\ddot{f}(x) = 6x - 6$



1. Put $\dot{f}(x) = 0$ $\therefore 3x^2 - 6x + 3 = 0$

$\therefore 3(x^2 - 2x + 1) = 0$ $\therefore 3(x-1)^2 = 0$ $\therefore x = 1$

, $\because \dot{f}(x) = 3(x-1)^2 > 0$ for all values of $x \in \mathbb{R} - \{1\}$

, $\because$ the function is continuous $\therefore$ The function is increasing on $\mathbb{R}$

, at $x = 1$ is neither local maximum nor local minimum point.

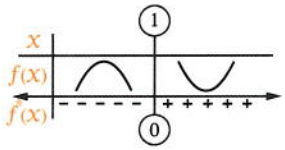
2. Put $\ddot{f}(x) = 0$ $\therefore 6x - 6 = 0$ $\therefore 6(x-1) = 0$ $\therefore x = 1$

$\therefore$ The curve of the function is convex upwards in $]-\infty , 1[$

, the curve of the function is convex downwards in $]1 , \infty[$

, there is an inflection point at $x = 1$, $\because f(1) = 8$

$\therefore$ The inflection point $(1 , 8)$



3. To find the intersection point with y-axis , put $x = 0$

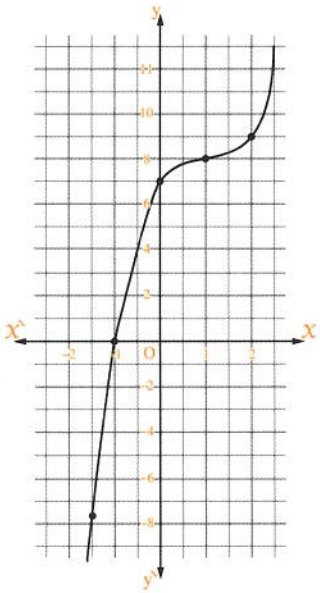
$\therefore f(0) = 7$

$\therefore$ The curve intersects the y-axis at $(0 , 7)$

4. Find some assistant points

$\therefore f(2) = 9$, $f(-1) = 0$, $f\left(-1\frac{1}{2}\right) = -7\frac{5}{8}$

$\therefore$ The curve passes through $(2 , 9)$, $(-1 , 0)$, $\left(-1\frac{1}{2} , -7\frac{5}{8}\right)$



5. Arrange all the previous points in the following table and represent them graphically.

x	$-1\frac{1}{2}$	-1	0	1	2
$f(x)$	$-7\frac{5}{8}$	0	7	8	9

Example 3

Sketch the curve of the function f given that :

1. f is a continuous function , its domain $[1 , 7]$, $f(1) = -1$, $f(5) = 4$, $f(7) = 1$

2. $\dot{f}(5) = 0$, $\dot{f}(x) > 0$ at $x < 5$, $\dot{f}(x) < 0$ at $x > 5$

3. $\ddot{f}(x) < 0$ at $1 < x < 7$

Solution

* Draw the perpendicular coordinate axes and plot the points $(1 , -1)$, $(5 , 4)$, $(7 , 1)$ and determine the domain as in figure (1)

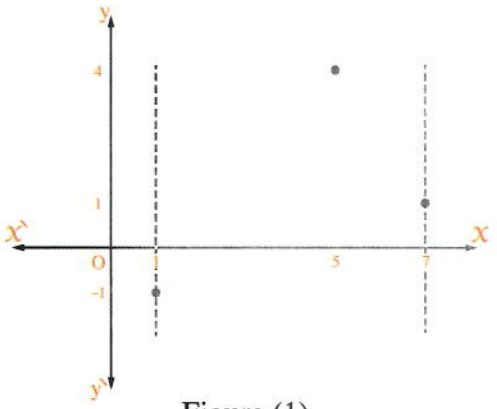


Figure (1)

* $\because \dot{f}(5) = 0$

, $\dot{f}(x)$ change its sign from positive to negative before and after $x = 5$

$\therefore f(5)$ = local maximum.

* $\because \ddot{f}(x) < 0$ (negative) for all values of $x \in$ the domain

$\therefore$ The curve is convex upwards

$\therefore$ Join the curve as in figure (2)

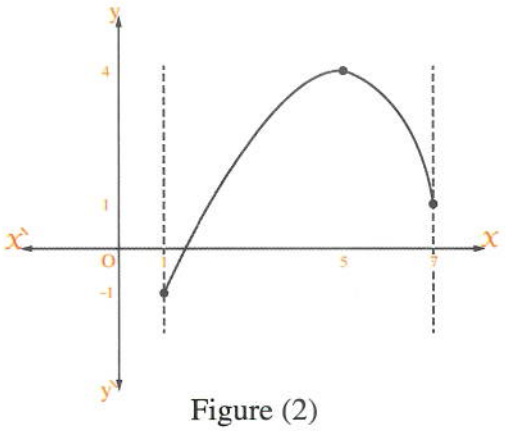
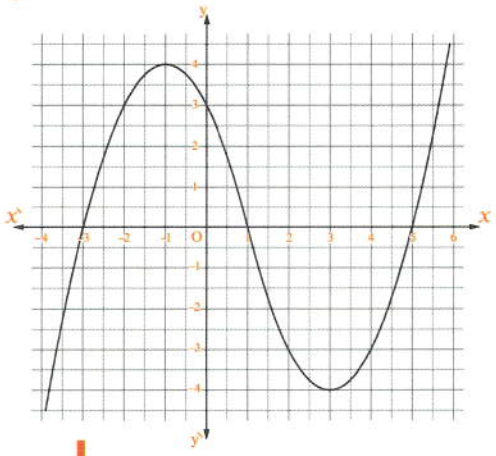


Figure (2)

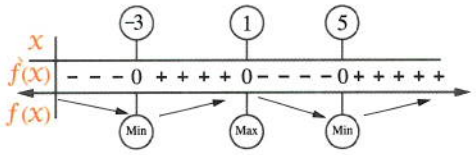
Remark 1

If the function represents the curve $f(x)$



- The point of local maximum is $(-1, 4)$
- The point of local minimum is $(3, -4)$
- The function is increasing in $]-\infty, -1[$, $]3, \infty[$
- The function is decreasing in $]-1, 3[$
- The inflection point is $(1, 0)$
- The curve is convex upwards in $]-\infty, 1[$
- The curve is convex downwards in $]1, \infty[$

If the curve represents the curve $\dot{f}(x)$



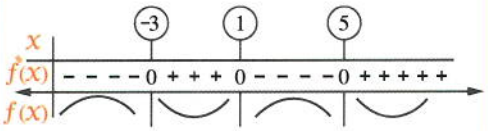
- There is local maximum of the function at $x = 1$
- There is local minimum of the function at $x = 5, x = -3$
- The function is increasing in $]-3, 1[$, $]5, \infty[$
- The function is decreasing in $]-\infty, 3[$, $]1, 5[$
- **Notice that:** The slope of the tangent of the curve $\dot{f}(x)$ is equivalent to $\ddot{f}(x)$

  , $\because$ the slope of the curve $\dot{f}(x) = 0$ at $x = -1, x = 3$

  $\therefore \ddot{f}(x) = 0$ at $x = -1, x = 3$ and the sign of $\ddot{f}(x)$ change around this values

  $\therefore$ There is an inflection point of the function f at $x = -1, x = 3$
- The curve is convex upwards in $]-1, 3[$
- The curve is convex downwards in $]-\infty, -1[$, $]3, \infty[$

If the curve represents the curve $\ddot{f}(x)$

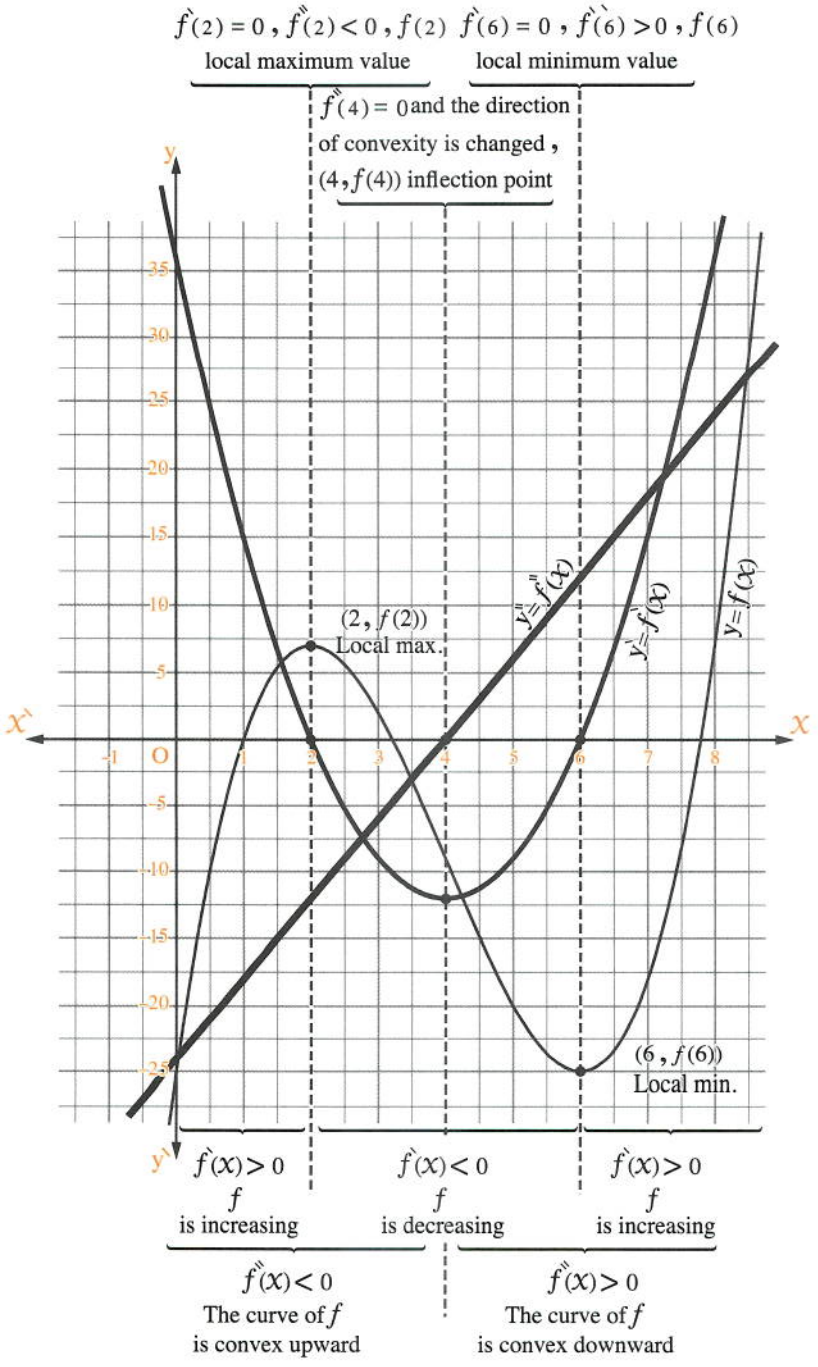


- The curve is convex upwards in $]-\infty, -3[$, $]1, 5[$
- The curve is convex downwards in $]-3, 1[$, $]5, \infty[$
- There are inflection points at $x = 5, x = 1, x = -3$

Remark 2

By studying the curve of the function f , let $f(x) = x^3 - 12x^2 + 36x - 25$, $f'(x) = 3x^2 - 24x + 36$, $f''(x) = 6x - 24$

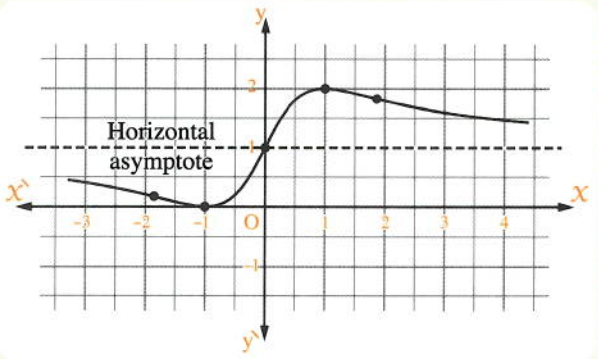
The following figure represents the relation between f , f' , f'' showing the local maximum and minimum points , the monotony , convexity and inflection points.



Activity

Some functions are difficult to draw its curves so you can use Ge Gebra or graph calculator or any other graphing program.

Draw the curve of the function and studying its properties.



For example :

The opposite figure represents the curve of the function f where $f(x) = \frac{(x+1)^2}{x^2+1}$

Notice that:

1. The function decreasing in $]-\infty, -1[$, $]1, \infty[$ and increasing in $]-1, 1[$

2. at $x = -1$ there is a local minimum

$= f(-1) = 0$, at $x = 1$ there is a local maximum $= f(1) = 2$

3. The curve is convex upwards in $]-\infty, -\sqrt{3}[$, $]0, \sqrt{3}[$ and convex downwards in $]-\sqrt{3}, 0[$, $]\sqrt{3}, \infty[$

4. at $x = -\sqrt{3}$, $x = 0$, $x = \sqrt{3}$ there are inflection points.

5. $\because \lim_{|x| \to \infty} \frac{(x+2)^2}{x^2+1} = 1$

$\therefore$ The curve has asymptote horizontal line whose equation "y = 1"

i.e. The curve approaches to the straight line "y = 1" with its two ends.

Remark

The horizontal asymptote line to the curve of the function f , has equation "y = a" where $a = \lim_{|x| \to \infty} f(x)$